

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Magnetic Effects of Electric Current

CBSE Science 2026-27 | Board-examinable

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

How can the direction of the magnetic field around a straight current-carrying wire be found?

Answer:

Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire.

Question 2

What happens to a current-carrying conductor placed in a magnetic field?

Answer:

It experiences a force when its current is not parallel to the magnetic field. The force direction depends on field and current directions and can be found by Fleming's left-hand rule.

Question 3

Why is a fuse placed in the live wire of a domestic circuit?

Answer:

A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk. In the live wire it disconnects the appliance from the supply.

Question 4

What is the central idea or skill tested in Magnetic Effects of Electric Current?

Answer:

The central focus is: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on Magnetic Effects of Electric Current.

Answer:

First point: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Second point: It experiences a force when its current is not parallel to the magnetic field. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in Magnetic Effects of Electric Current lead to an important result or conclusion?

Answer:

Begin with the key idea: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Then connect it to the result: It experiences a force when its current is not parallel to the magnetic field.

Magnetic Effects of Electric Current - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in Magnetic Effects of Electric Current?

Answer:

Use this supporting detail: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in Magnetic Effects of Electric Current.

Answer:

It experiences a force when its current is not parallel to the magnetic field. The force direction depends on field and current directions and can be found by Fleming's left-hand rule. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from Magnetic Effects of Electric Current?

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk.

Question 10

Compare the first and second important ideas in Magnetic Effects of Electric Current.

Answer:

The first idea is: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. The second idea is: It experiences a force when its current is not parallel to the magnetic field. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from Magnetic Effects of Electric Current.

Answer:

Cause or starting point: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Effect or outcome: It experiences a force when its current is not parallel to the magnetic field. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of Magnetic Effects of Electric Current?

Answer:

A clear sequence is: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Next, It experiences a force when its current is not parallel to the magnetic field. Finally, A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk.

Magnetic Effects of Electric Current - Continued

Question 13

Give a 40-50 word answer outline for an important question from Magnetic Effects of Electric Current.

Answer:

Start with the direct answer: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Add one supporting point: It experiences a force when its current is not parallel to the magnetic field. End with the outcome or message: A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in Magnetic Effects of Electric Current?

Answer:

Use an accurate scientific statement: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, Magnetic Effects of Electric Current?

Answer:

Connect the idea to its reason or result. For instance: It experiences a force when its current is not parallel to the magnetic field. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in Magnetic Effects of Electric Current.

Answer:

A sound inference is that a fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in Magnetic Effects of Electric Current.

Answer:

Apply the concept by first stating: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Then justify the response with: It experiences a force when its current is not parallel to the magnetic field.

Magnetic Effects of Electric Current - Continued

Question 18

What would make an answer from Magnetic Effects of Electric Current incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Also add: A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk.

Question 19

Prepare an Assertion-Reason style explanation based on Magnetic Effects of Electric Current.

Answer:

Assertion: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Reason: It experiences a force when its current is not parallel to the magnetic field. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from Magnetic Effects of Electric Current in two clear steps?

Answer:

Step 1: state the concept or event - Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Step 2: support it with the explanation or result - It experiences a force when its current is not parallel to the magnetic field.

Question 21

What answer structure is suitable for a 3-mark question from Magnetic Effects of Electric Current?

Answer:

Use three connected points: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. It experiences a force when its current is not parallel to the magnetic field. A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from Magnetic Effects of Electric Current?

Answer:

Give an introduction, develop these key points - Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. It experiences a force when its current is not parallel to the magnetic field. - and close with: A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk. Add a diagram, equation, example, or textual reference where relevant.

Magnetic Effects of Electric Current - Continued

Question 23

How can the ideas in Magnetic Effects of Electric Current be applied to a new situation or case study?

Answer:

Identify the relevant principle first: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Then use the chapter's reasoning to explain the case: It experiences a force when its current is not parallel to the magnetic field.

Question 24

Which detail from Magnetic Effects of Electric Current is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk. Explain it in context instead of using it as an isolated statement.

Question 25

Write a brief concluding statement for an answer on Magnetic Effects of Electric Current.

Answer:

A suitable conclusion is: A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk. This conclusion should follow naturally from the explanation rather than introduce a new point.

Question 26

Make a three-point revision note for Magnetic Effects of Electric Current.

Answer:

1. Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. 2. It experiences a force when its current is not parallel to the magnetic field. 3. A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from Magnetic Effects of Electric Current before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. - and supported it with a reason, example, calculation, or outcome such as: It experiences a force when its current is not parallel to the magnetic field.

Question 28

Why does Magnetic Effects of Electric Current remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. It experiences a force when its current is not parallel to the magnetic field. A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk.

Magnetic Effects of Electric Current - Continued

Question 29

Create a final 60-80 word answer plan for a long question from Magnetic Effects of Electric Current.

Answer:

Opening: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. Development: It experiences a force when its current is not parallel to the magnetic field. Conclusion: A fuse melts if excessive current flows, breaking the live connection and reducing fire or shock risk. Use complete sentences and include all required evidence, steps, or references.

Question 30

What is one last-minute revision question you should ask yourself about Magnetic Effects of Electric Current?

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: Use the right-hand thumb rule: point the right thumb in the current's direction; curled fingers show the magnetic field's direction around the wire. and It experiences a force when its current is not parallel to the magnetic field.

Sources: [CBSE Science curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

ONE SHOT MASTERY